Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/05/2023
PI Name:	Urbaniak, Camilla Ph.D.		
Project Title:	Microgravity Analogs as Proxies for Spaceflight to Validate Biofilm Formation and the Exchange of Genes Between Organisms		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	camilla.urbaniak@jpl.nasa.gov	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	626-840-6992
Organization Name:	ZIN Technologies, Inc		
PI Address 1:	4800 Oak Grove Dr		
PI Address 2:			
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:	NOTE: PI had NASA Postdoctoral Program Fellowship at NASA Jet Propulsion Laboratory until 2019; now with Zin Technologies, Inc., Middleburg, OH, but remains physically at NASA JPL address.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	01/04/2021	**End Date:**	01/03/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dr. Laurent Bentolila, UCLA, has been added as a collaborator for his expertise in confocal microscopy.		
COI Name (Institution):	Venkateswaran, Kasthuri Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC21K0320		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbial adaptation to space is an important area to study as it provides information on how microbes on spacecraft and the microbiome of astronauts will change when exposed to the space environment and how that may impact spacecraft integrity and function, and the health of astronauts. However, due to time and cost constraints of conducting studies on the International Space Station (ISS), scientists are not able to perform as many insightful space biology studies as they would like. For this reason, scientists use microgravity simulators to conduct space biology studies, but on Earth. However, there are very limited studies testing how these simulators compare to the "real" thing and the jury is still out on whether microgravity simulators are actually comparable to experiments performed on the ISS. For this reason, we wish to carry out a study that looks at how three simulated microgravity chambers, commonly used on the ground, compare to what would be observed on the ISS. This study would have a ground and a flight component and would look at two bacterial properties that are significant for short and long duration space missions--biofilm formation and the exchange of genetic material between bacteria.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	One of NASA's goals is to minimize the health risks associated with extended spaceflight, so it is critical that methods for preventing and treating spaceflight-induced illnesses be developed before astronauts embark upon long-duration space missions. In order for this to happen, it is important to learn how bacteria that play a role in human health and disease are affected by spaceflight, such as biofilm formation and horizontal gene transfer. Comparing these responses to popular and frequently used simulated microgravity devices on Earth, and verifying whether the same outcomes can be achieved, will allow more researchers to partake in space related microbiology and human microbiome work, without the extensive limitations imposed by conducting experiments on the ISS. This ultimately diversifies the field and opens up opportunities for scientists from all backgrounds. In addition, understanding microbial adaptation in the new and unique environment of space may help guide future research questions and studies into microbial adaptations on Earth under changing conditions, such as climate change.
Task Progress:	The following has been accomplished during the third year of this award: o Pre-flight testing review, completed February 2023. o Regular bi-weekly meetings between mission scientist, Principal Investigator (PI), payload developer, and other members of team. o Various hardware options have been sent by payload developer to the PI lab to test their usefulness for in-flight experiments. o OpNom registered and mission patch designed. Objective 1 (Horizontal Gene Transfer): • Ground tests completed with the Fluid Processing Apparatus / Group Activation Pack (FPA/GAP) flight hardware. • Science Verification Testing (SVT) started; in progress. Objective 2 (Biofilm Formation): • Various hardware designs have been tested, but none have, of yet, met the needs of the science. Hardware testing is still in progress. • Most protocols have been validated, though some are still ongoing.
Bibliography Type:	Description: (Last Updated: 11/01/2024)
Abstracts for Journals and Proceedings	Chen J, Schubert W, Venkateswaran K, Bentolila L, Urbaniak C. "ODYSSEY Study: Focusing on hardware development and ground protocols for a successful spaceflight. " 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Urbaniak C. "Microbial applications for sustainable space exploration beyond low Earth orbit." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Articles in Peer-reviewed Journals	Koehle AP, Brumwell SL, Seto EP, Lynch AM, Urbaniak C. "Microbial applications for sustainable space exploration beyond low Earth orbit." NPJ Microgravity. 2023 Jun 21;9(1):47. https://doi.org/10.1038/s41526-023-00285-0 ; PMID: 37344487; PMCID: PMC10284894 , Jun-2023
Articles in Peer-reviewed Journals	Santomartino R, Averesch NJH, Bhuiyan M, Cockell CS, Colangelo J, Gumulya Y, Lehner B, Lopez-Ayala I, McMahon S, Mohanty A, Santa Maria SR, Urbaniak C, Volger R, Yang J, Zea L. "Toward sustainable space exploration: A roadmap for harnessing the power of microorganisms." Nat Commun. 2023 Mar 21;14(1):1391. https://doi.org/10.1038/s41467-023-37070-2 ; PMID: 36944638; PMCID: PMC10030976 , Mar-2023
Articles in Peer-reviewed Journals	Urbaniak C, Grams T, Mason CE, Venkateswaran K. "Simulated Microgravity Promotes Horizontal Gene Transfer of Antimicrobial Resistance Genes between Bacterial Genera in the Absence of Antibiotic Selective Pressure." Life (Basel). 2021 Sep 13;11(9):960. doi: 10.3390/life11090960. PMID: 34575109; PMCID: PMC8468678. , Sep-2021